MASTER EXECUTIVE & TECHNICAL SYNTHESIS

AssessClear: Unified Pre-Pilot Synthesis, TabFM Analytics & Strategic Action Blueprint

Comprehensive Cross-Synthesis of Diagnostic Intelligence, Qwen 8-Report Evaluation, and Google TabFM Empirical Analysis

SYNTHESIS SCOPE	ANALYZED COHORT	DISTRICTS EVALUATED	READINESS VERDICT
3 Core Evaluation Documents	9 Schools · ~351 Student Copies	Katni, Indore, Bhopal, Raisen (MP)	Requires Attention (Conditional Green)

1. Executive Synthesis & Strategic Readiness Verdict

This master summary consolidates the three comprehensive pre-pilot evaluation documents: **(1) Diagnostic Intelligence & Strategic Action Plan**, **(2) Qwen 8-Report Evaluation**, and **(3) Google TabFM Empirical Data Analysis**.

Unified Strategic Verdict: REQUIRES ATTENTION BEFORE PILOT (Conditional Green Light)
The pedagogical hypothesis of AssessClear is firmly validated: multi-tiered rubric grading (1.0, 0.75, 0.40, 0.0) and distractor-misconception mapping provide immense diagnostic value over legacy binary grading. However, three critical defect clusters—(a) OCR/vision segmentation dropping Q1/Q2, (b) the "Double Effort" 80–90 min human review bottleneck, and (c) total failure in zero-connectivity schools—must be remediated to prevent teacher rejection during state rollout.

Major Validated Strengths

- **Constructive 3-Part Remediation:** Hand-holds students with *"What you did well, Think about this, Next steps"*.
- **Partial Attempt Credit:** Effectively scores partial mathematical approaches (0.40 / 0.75 weights).
- **Hindi OCR Capability:** Directly meets the language requirements of MP government schools.
- **Frugal Compute Cost:** Validated at ~₹0.12 per evaluation on Gemini-3.1-Flash-Lite.

Critical System Deficiencies

- **Question Dropping (Q1/Q2):** System dropped questions on 10–30% of sheets (e.g. 10 attempted → 7 evaluated).
- **Digit & Pencil Errors:** Misread handwriting (9996 → 3226); ignored pencil corrections.
- **Teacher Time Burden:** 183.4s mean total time/sheet results in >2 hrs of review per 40-student class.
- **Data Pipeline Truncation:** 200-evaluation database cap caused 59.3% data loss in Katni.

2. Unified District Evidence Matrix (Katni, Indore, Bhopal, Raisen)

District / Region	Schools & Sample	Key Field Strengths	Primary Critical Defects Logged	TabFM Risk Tier
Indore (Nandbag & Subhash Nagar)	2 Schools 100 Students (60 Math, 40 Hindi) DIET Observer	• Rapid batch upload efficient (≤10 sheets). • Fast 4G processing (22.8s/sheet). • Constructive feedback validated by DIET.	• Q1 & Q2 skipped on first page. • Rigid logic marked "3 months" wrong for "90 days". • Incorrect arithmetic box sum marked correct. • Dashboard omitted Hindi & Grade 7–8 data.	HIGH RISK (Q-drop)
Bhopal & Raisen (5 Schools: MS Green Park, MS Putlighar, HS Kanya in Bhopal; MS Kharbai & MS Bishankheda in Raisen)	5 Schools ~160 Students Grades 6–8 (Math & Hindi)	• Direct camera scanning saved local storage. • Structured feedback easy to grasp. • Hindi language relevance confirmed.	• Zero network in MS Kharbai halted app completely (0s latency), while MS Bishankheda processed normally (~26s). • Reviewing 30+ students individual feedback took >80 min. • Downloading individual PDFs fills phone storage. • Feedback focused on "what" rather than "how".	BLOCKER (No Net)
Katni (EPES Purwar & MS Devri Hatai)	2 Schools 91 Students (32 Purwar, 59 Hatai) Grade 6 Math/Hindi	• Step-by-step remedial handholding praised. • Partial attempt recognition encouraged effort.	• Dashboard showed 29 evaluations for 3 copies. • Session codes lost without manual paper notes. • Digit misrecognition (9996 → 3226). • 200-evaluation cap purged 59.3% of test history.	HIGH RISK (Data loss)

3. User Experience (UX) & The "Double Effort" Dilemma

The user experience journey across pre-pilot deployments revealed a significant operational gap between the tool's intended speed and the teacher's actual classroom reality:

Journey Stage	Designed Ideal Experience	Observed Pre-Pilot Reality	Operational Friction Level
1. Setup & Entry	Log in via Employee ID; select school; auto-resume session.	Session codes not saved in user profile; evaluators lost work if codes were not jotted down manually.	HIGH FRICTION
2. Ingestion	Rapid batch upload (up to 10 sheets) or 1-by-1 scanning.	Rapid batch works well on single pages; in rural areas (MS Kharbai), zero-network halted all uploads.	CRITICAL BLOCKER
3. AI Processing	Instant diagnostic scoring in seconds via Gemini vision.	AI latency = 22–75 sec. Image auto-crop clipped top/bottom margins, dropping Q1 and Q2.	CRITICAL DEFECT
4. Review & Action	Quick review of competency scores and color bands.	**"Double Effort" Bottleneck:** Teachers spent 1–3.5 min/paper transcribing remarks onto physical copies (80–90 min per class).	SEVERE OVERHEAD
5. Export & Archive	Export dual Teacher & Student diagnostic PDF.	Generated PDF contained only teacher feedback; student diagnostic view was omitted.	MEDIUM FRICTION

4. Google TabFM Empirical Modeling & Time Regression Results

The **TabFM Machine Learning framework** was executed across the pre-pilot dataset ($N=351$), yielding key quantitative findings:

TabFM Classification Breakdown

- **50.7% Medium Risk:** Standard evaluations in fast 4G areas.
- **39.3% High Risk:** Affected by Q1/Q2 skipping, digit errors, or >3.5 min review latency.
- **10.0% Critical Blocker:** Zero-network lockouts (MS Kharbai).

TabFM Time Burden Regression

- **Current Time per Sheet:** 183.4 sec (38.9s AI + 144.5s Review).
- **Class of 30 Students:** 91.7 minutes (1.53 hours).
- **Class of 40 Students:** 122.3 minutes (2.04 hours).
- **With Class Summary UI:** Slashed to **23.3 minutes (81% saved)**.

5. Unified Strategic Prioritization Roadmap (P0, P1, P2, P3)

Pri.	Technical & Product Action	Target Root Cause & Evidence	Operational Metric Impact	Timeline
P0	**Fix Vision Auto-Crop & Segmentation**	Bounding-box margins clip top/bottom headers (Indore & Katni).	Lifts Question Detection from 89.6% to >99.0%.	Immediate (Week 1)
P0	**Implement Offline-First PWA Ingestion**	Zero-network stall in MS Kharbai (Raisen).	Eliminates 10% critical blocker failure rate in rural schools.	Immediate (Week 1)
P0	**Refactor Schema & Lift 200-Eval Cap**	Firestore tracks question IDs instead of student copies; drops data >200.	Eliminates 59.3% data loss rate observed in Katni.	Immediate (Week 1)
P0	**Deploy Image Blur Quality Gate**	AI hallucinates scores on blurry images instead of prompting re-take.	Prevents corrupt diagnostic data from entering the dashboard.	Immediate (Week 1)
P1	**Build Class-Level Summary Heatmap UI**	80–90 min 1-by-1 screen review overhead per class.	Slashes teacher screen time from 122 min to 23 min per 40 students.	Immediate (Week 2)
P1	**Add 1-Click Teacher Score Override**	Teachers cannot correct AI misclassifications (Indore/Katni).	Guarantees 100% diagnostic accuracy and protects teacher trust.	Immediate (Week 2)
P1	**Active Session Drawer in Profile**	Evaluators lose session codes and duplicate work.	Enables 1-tap session recovery; lifts recovery rate to >98%.	Immediate (Week 2)
P2	**Prompt Tuning for Semantic Tolerance**	AI rejected "3 months" when key specified "90 days".	Eliminates false-negative scoring on valid conceptual synonyms.	Pre-Pilot Polish
P2	**Deep Integration with Sakhi AI**	Manual pedagogical actioning of diagnostic gaps.	Auto-sends WhatsApp lesson plans for class red-band competencies.	During Pilot

6. Pilot KPI Monitoring Framework & Research Hypotheses

Evaluation Category	Key Performance Metric	Target Benchmark	Measurement Method
Diagnostic Accuracy	Question Detection Rate AI Scoring Agreement Rate	> 99.0% detection > 95.0% scoring agreement	Compare AI scores against blind expert teacher evaluations.
Operational Speed	AI Processing Latency Teacher Review Overhead	< 25 sec per sheet < 15 sec per student	In-app telemetry logs and field time-motion observations.
Field Reliability	Offline Queue Sync Fidelity Session Restoration Rate	100% data sync > 98.0% session recovery	Track background sync logs across low-connectivity clusters.
Teacher Engagement	Weekly Active Usage Sakhi AI Remedial Follow-up	> 75% weekly adoption > 60% remedial resource use	Dashboard analytics and Sakhi AI WhatsApp interaction telemetry.

7. Final Leadership Conclusion

AssessClear represents an **unprecedented educational diagnostic innovation** for government school systems, offering high-fidelity, multi-tiered formative feedback at an economical compute cost (~₹0.12 per evaluation).

The pre-pilot exercises across Madhya Pradesh have decisively proved the concept while uncovering clear, fixable technical defects. By executing the **P0 and P1 resolutions** outlined in this master blueprint—specifically fixing auto-crop boundaries, adding offline PWA caching, lifting the 200-eval database cap, and deploying the Class-Level Summary UI—project leadership will ensure a seamless, high-trust, and impactful pilot deployment across the state.